# Certified Analytical Bounds for Multi-Phase Hypersonic Reachable Sets

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**Abstract**

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The vehicle is a rigid body, not a point mass (Appendix A). Its state

$$x = \big(r, \theta, \phi,\ u, v, w,\ q_0, q_1, q_2, q_3,\ \omega_1, \omega_2, \omega_3\big) \in \mathcal{X} \subseteq \mathbb{R}^{13}, \tag{2.1}$$

augmented by mass, recession depth $s$ and modal state $(\eta, \dot{\eta})$ where the corresponding mode or coupling is active, carries geocentric position, body-frame velocity, attitude quaternion and body rates as components. The attitude coordinates are redundant: every admissible state satisfies

$$q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1, \tag{2.2}$$

so the thirteen coordinates carry twelve degrees of freedom. The control

$$c = \big(\vec{\delta},\ \vec{\tau}_{\text{rcs}},\ \varsigma\big) \in U, \tag{2.3}$$

collects flap deflections $\vec{\delta}$, reaction-control moments $\vec{\tau}_{\text{rcs}}$ and, in a propulsive mode, throttle $\varsigma$; $U \subset \mathbb{R}^{n_c}$ is compact.

list deflection, RCS, and throttle limits (one citation each). compactness follows.

Crucially the incidence angles $(\alpha, \beta)$ are *outputs* of the attitude solution (**??**), not commanded quantities: trim is not assumed solvable, and attitudes the vehicle cannot hold are excluded automatically rather than by fiat. The reachable set cannot contain a state reached only by an unattainable incidence.

elaborate

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# A Six-Degree-of-Freedom Equations of Motion

This appendix records the rigid-body dynamics the body of the paper treats as given. Nothing here is a contribution; the derivation is standard entry mechanics assembled once with a fixed set of frames and a fixed parametrization. Frame errors are the classic 6-DOF defect and are invisible in the assembled algebra, so the transformations are stated explicitly and used without exception.

cite sources

## A.1 State vector and frames

**Frames.** Four frames are used throughout.

- $\mathcal{I}$: Earth-centred inertial (ECI), non-rotating.
- $\mathcal{E}$: Earth-centred Earth-fixed (ECEF), rotating with the constant planetary rate $\vec{\omega}_E = \omega_E\, \hat{z}_{\mathcal{I}}$.
- $\mathcal{L}$: local geocentric frame at the vehicle, with unit vectors $(\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi)$ pointing radially outward, east, and north.
- $\mathcal{B}$: body frame, fixed at the instantaneous centre of mass, axes along the vehicle's principal design directions.

The attitude quaternion $\vec{q}$ carries $\mathcal{I} \to \mathcal{B}$; the geocentric angles $(\theta, \phi)$ carry $\mathcal{I} \to \mathcal{L}$.

The direction-cosine matrix built from $\vec{q}$,

$$\mathbf{R}_{B/I}(\vec{q}) = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1 q_2 + q_0 q_3) & 2(q_1 q_3 - q_0 q_2) \\ 2(q_1 q_2 - q_0 q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2 q_3 + q_0 q_1) \\ 2(q_1 q_3 + q_0 q_2) & 2(q_2 q_3 - q_0 q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix}, \tag{A.1}$$

is polynomial of degree two in $\vec{q}$, which is the property Appendix C exploits. The geocentric map is

$$\mathbf{R}_{L/I}(\theta, \phi) = \begin{pmatrix} \cos\phi\cos\theta & \cos\phi\sin\theta & \sin\phi \\ -\sin\theta & \cos\theta & 0 \\ -\sin\phi\cos\theta & -\sin\phi\sin\theta & \cos\phi \end{pmatrix}. \tag{A.2}$$

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